

CRISTOFF STAN

Senior Software Engineer

Location: Tallinn, Harjumaa, Estonia | Email: cristoffstan@gmail.com
LinkedIn: <https://www.linkedin.com/in/cristoff-stan-b53490362/> | Phone Number: +372 58619311

PROFESSIONAL SUMMARY

Seasoned Senior Software Developer with over 9 years of experience in distributed systems, real-time data streaming, and large-scale microservices architectures. Specialized in Apache Kafka ecosystem, including stream processing, topic design, access management, performance tuning, and infrastructure operations. Adept in translating business requirements into resilient event-driven architectures. Proven experience with CI/CD pipelines, Kafka security, monitoring, and recovery strategies. Passionate about delivering high-throughput, low-latency solutions across B2B ecosystems.

TECHNICAL SKILLS

- **Kafka Ecosystem:** Kafka Streams, Kafka Connect, Schema Registry, Zookeeper, MirrorMaker 2.0
- **Backend:** Java, Kotlin, Node.js, C#, .NET Core
- **Cloud & DevOps:** Docker, Kubernetes, Azure, AWS, Terraform, Azure DevOps, GitHub Actions
- **Databases:** PostgreSQL, MongoDB, Redis, Cassandra
- **Monitoring & Observability:** Prometheus, Grafana, Splunk, Elastic Stack
- **Security & Access:** OAuth2, TLS, RBAC, Kerberos, Confluent RBAC
- **Data Management:** ETL pipelines, Data Purging, TTL, Archiving
- **Others:** Git, REST APIs, Agile, Scrum, CI/CD, Jira

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Led Kafka-based data pipeline initiatives supporting real-time data ingestion, schema validation, and transformation workflows for mission-critical applications.
- Architected Kafka stream processors that enabled microservices to decouple and scale asynchronously across product lines with failover strategies.
- Conducted performance tuning, benchmarking throughput and latency using synthetic loads and real-world traffic for production optimization.
- Automated Kafka topic creation, ACL assignments, and partition scaling using CI/CD pipelines with Terraform and Confluent CLI.
- Designed disaster recovery and high-availability Kafka clusters across regions using MirrorMaker 2.0 and monitored replication lags.

- Collaborated with InfoSec and SRE teams to implement TLS, SASL, and role-based access control across Kafka producers and consumers.
- Analyzed logs, handled incident resolution, and led postmortem reporting for Kafka-related production failures and anomalies.
- Contributed to capacity planning, forecasting cluster growth trends using retention metrics, broker CPU/memory, and consumer lag analytics.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Migrated legacy data ingestion system to a Kafka-first event streaming platform reducing ETL latency from hours to seconds.
- Created reusable Kafka templates for producer/consumer services using Spring Boot and Avro schemas with strict schema evolution checks.
- Built monitoring dashboards and alerting pipelines for Kafka brokers using Grafana, Prometheus, and custom exporter metrics.
- Provided production support by diagnosing consumer lags, topic skew, offset issues, and corrupted messages post-release.
- Introduced TTL-based purging policies and compacted topics to support GDPR compliance and reduce disk usage across retention windows.
- Wrote internal documentation, guidelines, and Terraform modules for Kafka topic lifecycle automation across environments.
- Handled incident management, root cause analysis, and rollback strategies involving Kafka version upgrades and configuration errors.
- Engaged with application teams to audit access, optimize consumer groups, and improve parallelism using partition strategies.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Contributed to Kafka integration across backend microservices for customer notification, analytics, and data archival modules.
- Designed fault-tolerant producers and retry mechanisms for time-sensitive financial transaction data pipelines.
- Implemented Kafka change data capture (CDC) patterns using Debezium connectors for PostgreSQL to enable near real-time syncs.
- Reviewed and approved Kafka PRs including schema changes, topic configurations, and broker tuning for various development squads.
- Collaborated with QA and DevOps to simulate load tests and throttle Kafka traffic during major release cycles.
- Audited existing topics and aligned them with business domains, ensuring naming conventions, partition counts, and replication factors were standardized.
- Built dashboards with Elasticsearch and Kibana for end-to-end Kafka message tracking and consumer throughput visibility.
- Helped orchestrate Kafka version upgrades with zero-downtime cutovers using blue/green deployment strategies.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed Kafka-based microservices to integrate CRM and email systems, enabling asynchronous communication and real-time data synchronization across multiple business platforms.
- Designed and implemented Kafka topics with well-defined partitioning and replication configurations to ensure high availability and fault tolerance.
- Created custom Protobuf serializers and deserializers to optimize message payloads, enforce schema consistency, and reduce payload size for cross-service communication.
- Configured Kafka Connect to import/export records from CRM databases to Kafka topics, supporting downstream processing with CDC patterns.
- Participated in topic naming strategy workshops and contributed to a reusable topic configuration guide based on Confluent best practices.
- Integrated Kafka consumer groups into email dispatch workflows to decouple trigger events from mail service execution and support load balancing.
- Assisted in implementing stream processors using Kafka Streams API to aggregate customer engagement data for marketing analytics.
- Developed internal testing suites for Kafka pipelines, using embedded Kafka clusters to validate consumer/producer logic, schema validation, and error handling.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Built visualization dashboards for Kafka topic metrics using JMX exporters and open-source tools like Grafana and Prometheus to monitor broker and consumer performance.
- Assisted in Kafka cluster setup and configuration for internal PoC projects, including broker replication, log retention settings, and topic cleanup policies.
- Contributed to a prototype that used Kafka as a central messaging hub for IoT sensor data, implementing a minimal viable stream ingestion and processing pipeline.
- Developed Python scripts to simulate producers sending sensor data into Kafka topics and tested consumer group behavior under high ingestion rates.
- Created reports on Kafka consumer lag analysis and topic throughput for weekly performance reviews, aiding in early capacity planning.
- Implemented data archiving routines using Kafka Connect and sink connectors to persist stream data to long-term object storage.
- Evaluated different Kafka serialization formats including Avro and JSON, documenting performance trade-offs and schema management requirements.
- Participated in cross-functional design reviews to align Kafka topic structures with business domains and proposed retention strategies based on data criticality.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- **Confluent Certified Developer for Apache Kafka (CCDAK)** – 2024
 - **Kafka Streams and ksqlDB Essentials** – Udemy, 2023
 - **Designing Event-Driven Microservices** – O'Reilly, 2023
 - **AWS Certified Solutions Architect – Associate** – 2022
 - **Azure Fundamentals (AZ-900)** – 2022
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PROJECTS

-  **EventBus++** — *Kafka Multi-Region Stream Relay*
Built a Kafka event bus solution with MirrorMaker 2.0 supporting bidirectional message replication between Europe and US regions, with fault-tolerant stream restoration and custom audit logging.
-  **StreamGuard** — *Kafka Monitoring & Auditing Toolkit*
Developed a Python-based CLI and Grafana dashboards that collect and visualize topic retention, partition skew, ACL mismatches, and consumer group anomalies in real-time.
-  **SecureKafkaMesh** — *Kafka Access Governance Platform*
Led implementation of a Kafka security layer to manage topic-level RBAC using Confluent APIs and Keycloak integration for automated ACL provisioning via CI/CD pipelines.